

Customer Churn Analysis Project Report

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This report presents a business-friendly overview of the Customer Churn Analysis project. The objective of the project was to identify factors that contribute to customer attrition, quantify business risk, and provide actionable recommendations for improving customer retention. **Key Dataset**

Statistics

- Total Customers: 7,043
- Churned Customers: 1,869
- Churn Rate: 26.54%

Project Objectives

- Understand customer behavior and churn patterns.
- Identify high-risk customer segments.
- Estimate revenue impact of customer churn.
- Build an interactive Power BI dashboard for decision-makers.
- Provide recommendations to reduce churn and improve retention.

Technology Stack

Python: Used for data preparation, cleaning, and exploratory analysis.

Pandas: Data transformation, aggregation, and feature preparation.

NumPy: Numerical computations and data processing support.

Matplotlib & Seaborn: Exploratory visualizations and trend analysis.

SQL: Querying, filtering, aggregation, and business metric extraction.

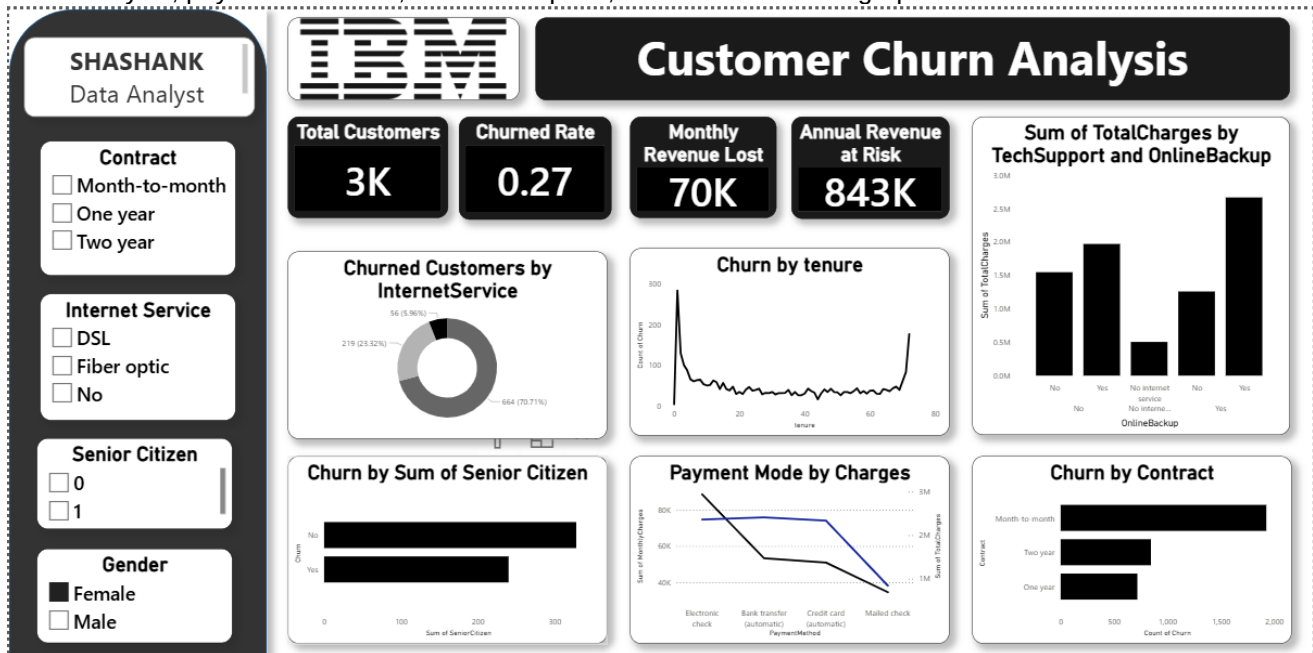
Power BI: Dashboard development, KPI tracking, and interactive reporting.

Project Workflow

1. Data Collection from the Telco Customer Churn dataset.
2. Data Cleaning and preprocessing using Python and Pandas.
3. Exploratory Data Analysis using Matplotlib and Seaborn.
4. SQL-based querying and metric validation.
5. Dashboard design and KPI creation in Power BI.
6. Business interpretation and recommendation generation.

Power BI Dashboard Overview

The dashboard provides an executive-level overview of churn behavior, revenue risk, contract analysis, payment behavior, service adoption, and customer demographics.



Main KPIs Displayed

- Total Customers
- Churn Rate
- Monthly Revenue Lost
- Annual Revenue at Risk
- Churn by Contract Type
- Churn by Tenure
- Churn by Payment Method
- Service-Level Analysis

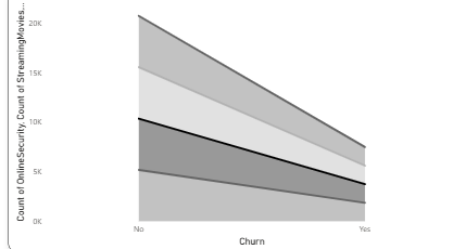
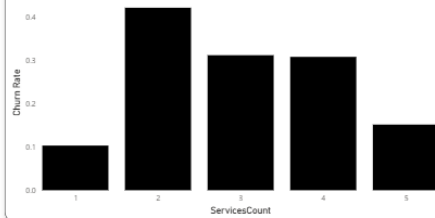
Deep Dive Analysis & Recommendations

Deep Dive & Recommendations

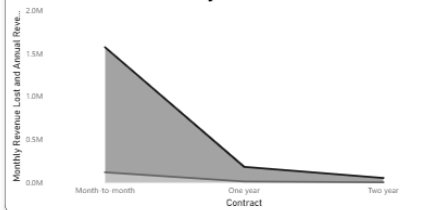
Recommendations

- Convert month-to-month customers to annual contracts with loyalty incentives.
- Launch a 90-day onboarding program for new customers.
- Investigate fiber optic service quality high churn despite premium pricing. Incentivize auto-pay enrollment reduces churn by 30 percentage point .

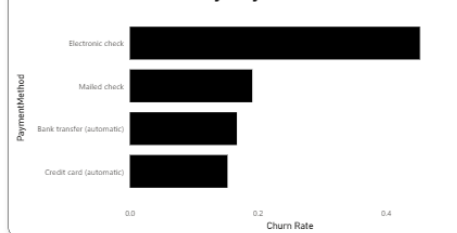
Churn Rate by ServicesCount



Monthly Revenue Lost and Annual Revenue At Risk by Contract



Churn Rate by PaymentMethod



Key Business Insights

- Month-to-month customers exhibit the highest churn behavior.
- Fiber optic customers show relatively high churn despite premium services.
- Electronic check users have elevated churn rates compared to auto-payment users.
- New customers are more likely to leave during the early tenure period.
- Customers using more services tend to have different churn patterns that can guide targeted retention efforts.

Recommendations

- Encourage migration from month-to-month plans to annual contracts.
- Launch onboarding and engagement programs for new customers.
- Improve customer support experience for high-risk segments.
- Promote automatic payment methods through incentives.
- Monitor service quality and satisfaction among fiber optic customers.
- Use proactive retention campaigns for customers showing churn signals.

Business Impact

This solution helps organizations identify revenue leakage caused by customer attrition, prioritize retention efforts, improve customer lifetime value, and support data-driven decision making. The dashboard enables stakeholders from non-technical backgrounds to monitor churn trends through intuitive visualizations and actionable KPIs.

Conclusion

The Customer Churn Analysis project combines Python, SQL, and Power BI to transform raw customer data into actionable business insights. The resulting dashboard and recommendations provide a strong foundation for customer retention strategies and revenue protection initiatives.